

GUI
+
TKinter

PA – 18241

Programação
Aplicada

Ricardo Marau
marau@ua.pt



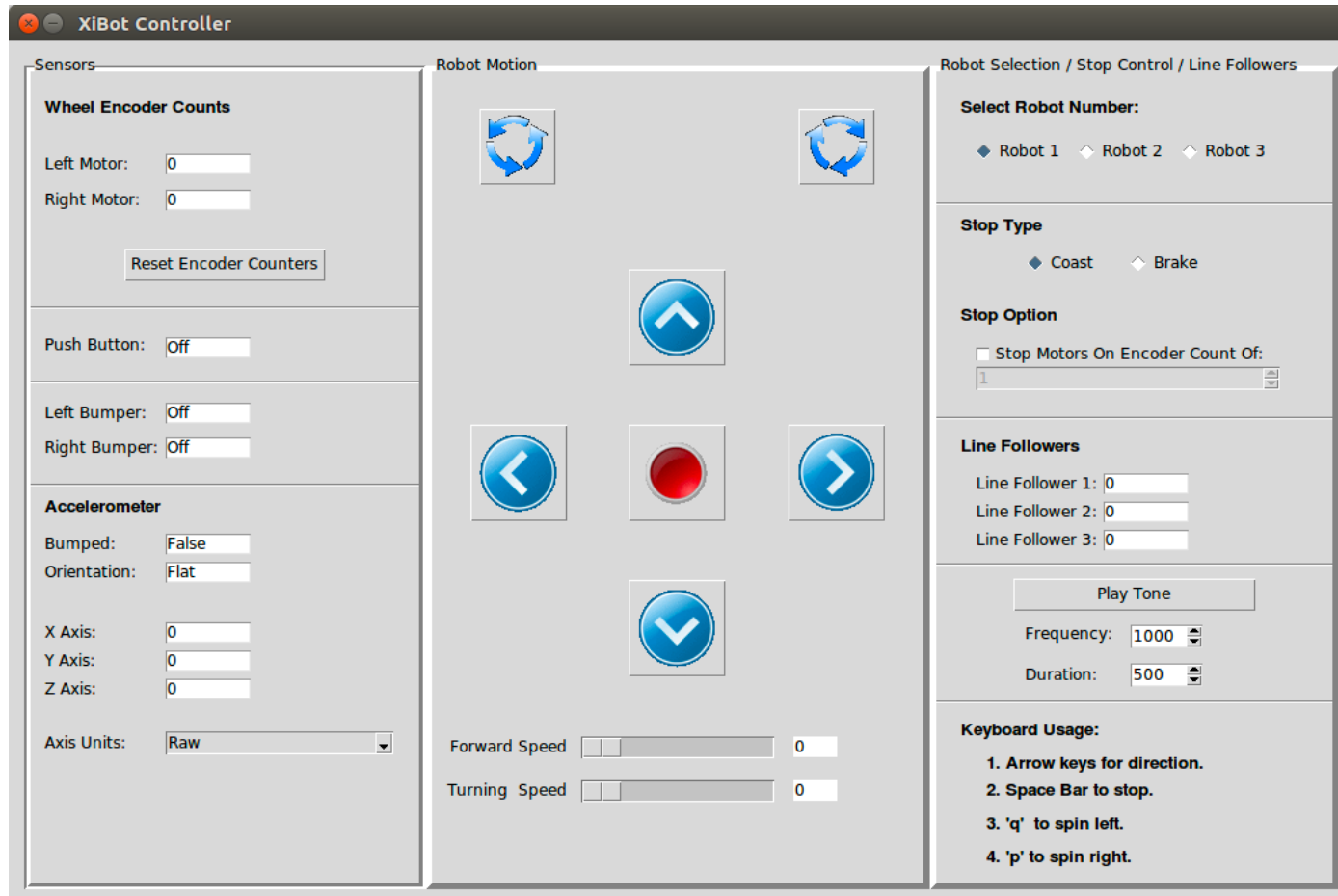
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GUI

- ◉ Graphical User Interface
 - ◉ Python:
 - ◉ **Tkinter** – standard library.
 - ◉ PyQt
 - ◉ PyGtk
 - ◉ ...
 - ◉ <https://wiki.python.org/moin/GuiProgramming>

Tkinter



Widgets

- ◉ button
- ◉ canvas
- ◉ checkbutton
- ◉ combobox
- ◉ entry
- ◉ frame
- ◉ label
- ◉ labelframe
- ◉ listbox
- ◉ menu
- ◉ menubutton
- ◉ message
- ◉ notebook
- ◉ tk_optionMenu
- ◉ panedwindow
- ◉ progressbar
- ◉ radiobutton
- ◉ scale
- ◉ scrollbar
- ◉ separator
- ◉ sizegrip
- ◉ spinbox
- ◉ text
- ◉ treeview

```
import tkinter
import tkinter.messagebox
```

```
top = tkinter.Tk()
```

```
def helloCallBack():
    tkinter.messagebox.showinfo( "Hello Python", "Hello World")
```

```
B = tkinter.Button(top, text = "Hello", command = helloCallBack)
```

```
B.pack()
top.mainloop()
```

```
top = tkinter.Tk()
top.geometry("800x600")
top.title("Janela de POO")
```



```
import tkinter
import tkinter.messagebox
```

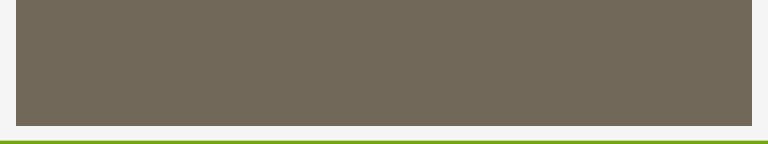
```
top = tkinter.Tk()
#top.geometry("800x600")
```

```
def helloCallBack():
    tkinter.messagebox.showinfo( "Hello Python", "Hello World")
```

```
B = tkinter.Button(top, text ="Hello", command = helloCallBack)
```

```
B.pack()
top.mainloop()
```





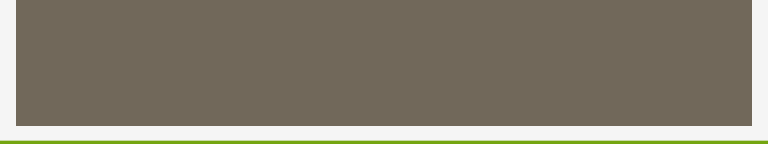
```
import tkinter
import tkinter.messagebox
```

```
top = tkinter.Tk()
top.geometry("800x600")
top.title("Janela de POO")
```

```
def helloCallBack():
    tkinter.messagebox.showinfo( "Hello Python", "Hello World")
```

```
B = tkinter.Button(top, text = "Hello", command = helloCallBack)
L = tkinter.Label(top, text = "Label", bg="black", fg="white" )
```

```
L.pack()
B.pack()
top.mainloop()
```



```
import tkinter
```

```
top = tkinter.Tk()  
top.geometry("800x600")  
top.title("Janela de POO")
```

```
def toUpperCaseCallBack():  
    L['text'] = E.get().upper()
```

```
B = tkinter.Button(top, text = "Hello", command = toUpperCaseCallBack)  
E = tkinter.Entry(top, bg="black", fg="white")  
L = tkinter.Label(top, text = "Label" )
```

```
E.pack()  
B.pack()  
L.pack()
```

```
top.mainloop()  
quit()
```

```
class EntryPack():
```

```
    def __init__(self, parent, comando):
```

```
        self.frame = tkinter.Frame(parent)
```

```
        self.button = tkinter.Button(self.frame, text="Submete", command=
```

```
        self.entry = tkinter.Entry(self.frame, bg="black", fg="white")
```

```
        self.entry.pack(side=tkinter.LEFT) #Alinhado à esquerda do anterior
```

```
        self.button.pack(side=tkinter.LEFT) #Alinhado à esquerda do anterior
```

```
        self.frame.pack()
```

```
    def getText(self):
```

```
        return self.entry.get()
```

```
top = tkinter.Tk()
```

```
def toUpperCaseCallBack():
```

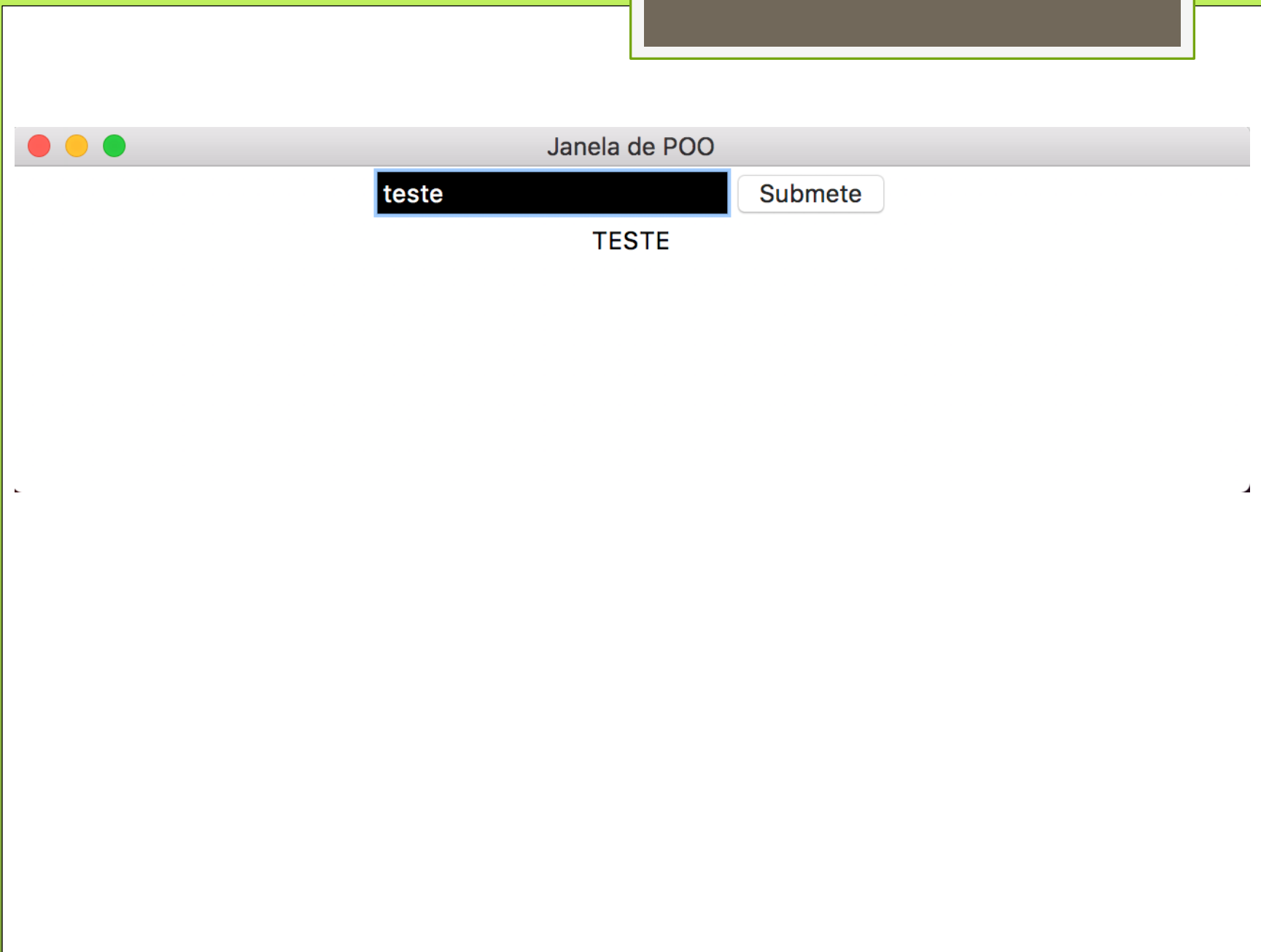
```
    L['text'] = entryPack.getText().upper()
```

```
entryPack = EntryPack(top,toUpperCaseCallBack)
```

```
L = tkinter.Label(top, text = "Label" )
```

```
L.pack()
```

```
top.mainloop()
```



```

class EntryPack(tkinter.Frame):
    def __init__(self, parent, comando):
        super().__init__(parent)
        self.button = tkinter.Button(self, text="Submete", command=lambda:comando(self) )
        self.entry = tkinter.Entry(self, bg="black", fg="white")
        self.entry.pack(side=tkinter.LEFT) #Alinhado à esquerda do anterior (parent)
        self.button.pack(side=tkinter.LEFT) #Alinhado à esquerda do anterior (entry)

    def getText(self):
        return self.entry.get()

top = tkinter.Tk()
def toUpperCaseCallBack(obj):
    L['text'] = obj.getText().upper()

entryPack = EntryPack(top,toUpperCaseCallBack)
L = tkinter.Label(top, text = "Label" )

entryPack2 = EntryPack(top,toUpperCaseCallBack)
entryPack.pack()
entryPack2.pack()
L.pack()

top.mainloop()

```



Questões?

- https://www.tutorialspoint.com/python/python_gui_programming.htm